WACV #3638

Crop-Label Alignment: Auditing Crop-Dependent Auto-Labels in Remote-Sensing Segmentation

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Abstract

Some automated segmentation pipelines fit context-dependent label rules inside each crop. Overlapping crops can then assign different labels to the same source pixel. Under patch-wise scoring, a model can be rewarded for following the crop-specific label instead of the scene.

We introduce crop-label alignment, κ (distinct from Cohen's kappa). On pixels whose crop-specific labels disagree, it measures whether predictions follow the label in the crop being read more often than labels from other covering crops. For any fixed crop-invariant predictor, $\kappa = 0$ algebraically. This removes the need to simulate a null; attributing a nonzero measurement to the training labels still requires a matched control. A per-pixel threshold yields $|\kappa| < 1.4 \times 10^{-15}$ in our implementation.

We calibrate the diagnostic on Sen1Floods11 (Sentinel-1, 11 flood events) using a dial from its published event threshold to thresholds fitted inside crops. On the fixed disagreement set (24.71% of covered pixels), the end-to-end contrast is +0.2613. After averaging seeds, the five levels are ordered in every held-out event; after averaging events, they are ordered at each of three training seeds. The same construction changes both the mean threshold and label quality. Its −0.29 expert-label mIoU difference is descriptive and does not isolate causal cost.

Surveys without trained models show that crop-adaptive SAR rules produce disagreement on 12–18% of covered pixels, while fixed and scene-fitted variants produce none. On external imagery, a published colour labeller applied per crop yields 37.04%; fitting its context once per scene yields zero.

On a 17-acquisition sea-ice benchmark assembled from that published pipeline, $P(\mathcal{A}) = 13.22\%$ and the optical-only primary model trained on per-patch labels yields $\kappa = +0.1518$, compared with +0.0097 for a matched scene-label control. The +0.1421 contrast is positive in all 17 dependent folds; tested code and outputs are released.

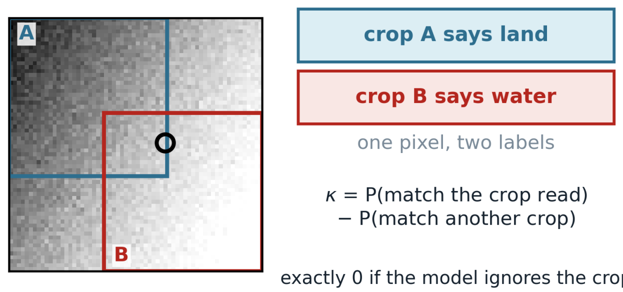

Figure 1. The failure this paper measures. A labeller run inside each crop computes a different threshold in each, here −21.5 and −15.5 dB, so a pixel between them comes out water in one window and land in the other.

1. Introduction

Auto-labelling made large segmentation benchmarks possible. It also made some of them circular, because the label can depend on which crop the model reads a pixel in (Fig. 1). We are not aware of a diagnostic tailored to this failure with an algebraically fixed reference value. Partial-input baselines answer the neighbouring question [13, 20], but diagnose a dataset by training a second model and reading a gap whose null must be estimated. What is needed here is an estimand whose value under crop-invariant predictions is known in advance and whose label-only component can be computed on a benchmark one did not build.

This paper supplies one, and applies it to two benchmarks. Foundation-model masks can be used to construct remote-sensing segmentation labels [37], and a benchmark that adopts them per crop inherits the property it tests for.

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Contributions.

- A directional statistic, κ, for how far a model's output tracks the crop it is reading, whose null is exact rather than estimated: $\kappa = 0$ for any crop-invariant predictor on the fixed evaluated data, with no simulated null calibration (Proposition 1). A matched contrast descriptively assesses whether excess alignment is associated with the training-label construction rather than an architecture-only floor (Sec. 3.3).
- A calibration on Sen1Floods11 in which we rebuild the published labeller with a dial, and a within-chip threshold scramble that preserves each chip's threshold multiset while attenuating, but not eliminating, the threshold–input correspondence (Sec. 6).
- A measurement on a 17-acquisition sea-ice benchmark we assembled from a published labeller, against a matched control trained on labels regenerated once per scene (Sec. 8).
- A survey of labelling rules on public imagery with nothing trained, including the published labeller run per crop on chips it never saw, separating the rules that survive being applied per crop from those that do not (Sec. 4.2, Sec. 8.1).
- A single-file implementation with a self-test, and a protocol: label once per scene, split by acquisition, score each source pixel once (Sec. 9).

Terms. A benchmark is crop-dependent when one source pixel can receive different labels depending on the crop it is read in. Computing labels from the model's own input is not sufficient: a rule applied pointwise over a whole scene reads those pixels and stays crop-invariant. Crop-label alignment (κ) is defined in Section 3.

2. Related work

Crop-dependent labelling sits between several literatures. The distinction this paper turns on is the one between shortcut learning and label noise.

Shortcut learning. Documented cases include hospital-specific tokens in chest radiographs [38], background rather than animal in wildlife recognition [4], and the general account of shortcut learning as a property of gradient descent rather than of any one dataset [12, 25]. In each case the cue was identified after a model had trained and generalised badly. Crop-label alignment is a forward test with a value known under the null, so it names the cue in advance.

Annotation artefacts and partial-input baselines. The closest analogues are partial-input baselines. Hypothesis-only models on natural language inference showed that labels were partly predictable from the annotation process, and that leaderboard progress had been measuring that predictability [14, 32]; vision does the same with question-only baselines [13, 20], whose licensed inference has itself been questioned [11]. The difference is that a crowdworker's habits cannot be recovered and an algorithmic labeller can: we invert one to its constant in 17 of 17 folds and recover another's threshold and kernel to within 0.25 dB on 11 of 11 events, which lets us build the artefact with a dial instead of only detecting it.

Relation to label noise. The standard frame for imperfect labels is noise, with a large literature on robust losses, noise-transition estimation, and the observation that many test labels in standard benchmarks are simply wrong [28, 29, 36]. Crop-dependent error sits at one end of that literature rather than outside it: input-independent noise supplies no stable input-predictive signal, whereas instance-dependent noise can, and a crop-dependent label is the limiting case, fixed by the crop the model is reading. A predictor can therefore align with it and be rewarded under patch scoring where input-independent noise is only penalised. The appropriate remedy is to generate labels outside the crop and evaluate at source-pixel resolution.

Weak and programmatic supervision. Labelling functions and foundation-model pseudo-labels are two mechanisms that can produce crop-dependent labels at scale [33, 37]. What matters is what the labelling function reads: a per-pixel rule survives being applied per crop, a rule with a threshold estimated from the crop does not. Benchmarks shipping expert labels beside algorithmic ones [2] would let κ and its association with expert-label performance be evaluated without constructing anything.

Overlapping windows and tiling artefacts. Cross-window consistency has been used as a training loss on overlapping predictions [10], and remote-sensing work documents boundary errors from tiled inference and stitching [15]. Recent reference-free seam metrics test whether prediction gradients at tile boundaries differ from surrounding content [8]. Those methods ask whether predictions agree across windows or form visible seams. $P(\mathcal{A})$ instead uses labels only, Ω uses prediction variation only, and κ conditions on label disagreement and signs that variation by which crop's label the prediction follows. Consequently seam or consistency metrics can fire when labels are crop-invariant (where κ is undefined), while $\kappa = 0$ can also hide cancelling signed effects. The diagnostic is therefore narrower and complements overlap consistency.

Benchmark statistics. A separate line argues that benchmark comparisons are reported without the variance or the power to support them, and that seed and data-order variation can exceed the differences claimed [6, 7, 9, 34]. We therefore treat held-out folds as descriptive evaluation units rather than independent replicates.

Spatial validation. Splitting spatially autocorrelated data at random inflates measured skill [22, 23, 31, 35], and treating correlated observations as independent replicates has a long history [16]; the leakage taxonomy places this among failures that survive peer review [21]. Our protocol repairs are the standard remedies applied to a benchmark that used none; what we add is a diagnostic for the label alignment that can remain after those choices.

3. Crop-label alignment

3.1. Definition

A labeller produces, for each crop c and each pixel p inside it, a label $L_c(p)$. When it reads only the pixels inside the crop, two crops covering the same source pixel can assign it different labels. A predictor f maps a pixel and the crop it is read in to a class, written $f(p \mid c)$.

For a source pixel p, let $C(p)$ be the set of evaluated crops covering it and supplying a valid label, and $m(p) = |C(p)|$. The artefact set is

$$\mathcal{A} = \{\, p : \exists\, c, c' \in C(p) \text{ with } L_c(p) \neq L_{c'}(p) \,\}. \tag{1}$$

Off $\mathcal{A}$ the labeller is locally single-valued and there is nothing to detect. A pixel covered by one crop can never enter $\mathcal{A}$, so $m(p) \geq 2$ throughout.

For $p \in \mathcal{A}$ and $c \in C(p)$, with $\mathbf{1}[\cdot]$ the indicator, define

$$\mathrm{own}(p, c) = \mathbf{1}[f(p \mid c) = L_c(p)]\,, \tag{2}$$

$$\mathrm{oth}(p, c) = \frac{1}{m(p) - 1} \sum_{c' \neq c} \mathbf{1}[f(p \mid c) = L_{c'}(p)]\,, \tag{3}$$

where here and below sums over c or c' run over $C(p)$. Equation (2) asks whether the prediction matches the label of the crop being read, Eq. (3) how often it matches another covering crop's label. Averaging those differences over source pixels,

$$\kappa = \frac{1}{|\mathcal{A}|} \sum_{p \in \mathcal{A}} \frac{1}{m(p)} \sum_{c \in C(p)} \big[\, \mathrm{own}(p, c) - \mathrm{oth}(p, c) \,\big]. \tag{4}$$

Each $p \in \mathcal{A}$ counts once. Averaging over (p, c) instances instead gives κ_{cr}, the crop-read weighting, in which a pixel counts once per covering crop; on $\mathcal{A}$, coverage runs from 2 to over 100, so the two differ. We report Eq. (4) throughout and κ_{cr} in the supplement. This statistic lies in $[-1, 1]$, is conditional on $p \in \mathcal{A}$, and is unrelated to Cohen's kappa despite the shared symbol. We therefore report its disagreement prevalence $P(\mathcal{A}) = |\mathcal{A}| / |\{p : m(p) > 0\}|$ over evaluation-valid covered source pixels alongside the primary benchmark results: κ describes alignment severity where labels conflict, not how often conflict occurs.

3.2. The null is exact

Proposition 1. Let $\mathcal{A}$ be non-empty and let f be crop-invariant on $\mathcal{A}$, meaning $f(p \mid c) = f(p)$ for every $p \in \mathcal{A}$ and every $c \in C(p)$. Then $\kappa = 0$ exactly.

Proof. Fix $p \in \mathcal{A}$, write $m = m(p)$ and $y = f(p)$, which by hypothesis does not depend on the crop. For each class k let $n_k = |\{c \in C(p) : L_c(p) = k\}|$, so that $\sum_k n_k = m$. Summing Eq. (2) over the crops covering p,

$$\sum_c \mathrm{own}(p, c) = \sum_c \mathbf{1}[y = L_c(p)] = n_y. \tag{5}$$

Summing Eq. (3) and exchanging the order of summation,

$$\begin{aligned} \sum_c \mathrm{oth}(p, c) &= \frac{1}{m - 1} \sum_c \sum_{c' \neq c} \mathbf{1}[y = L_{c'}(p)] \\ &= \frac{1}{m - 1} \sum_{c'} \mathbf{1}[y = L_{c'}(p)] \cdot (m - 1) \\ &= n_y, \end{aligned} \tag{6}$$

because each c' is counted once for every c other than itself, that is $m - 1$ times, and the normaliser cancels it.

So Eq. (5) and Eq. (6) agree at every $p \in \mathcal{A}$, and therefore

$$\sum_{c \in C(p)} \big[\mathrm{own}(p, c) - \mathrm{oth}(p, c)\big] = n_y - n_y = 0 \quad \text{for every } p \in \mathcal{A}. \tag{7}$$

Equation (4) is a weighted sum of that inner quantity, so it is 0; and because Eq. (7) holds pixel by pixel, any weighting over p gives 0, including κ_{cr}. $\square$

Nothing in the argument uses the number of classes, the label distribution, the coverage $m(p)$, or any property of the imagery; it needs only that $\mathcal{A}$ is non-empty, since κ averages over the source pixels in it. For fixed evaluated predictions, zero is an algebraic identity rather than a null distribution that must be simulated

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or estimated. This says nothing by itself about population uncertainty across scenes.

$\kappa \neq 0$ certifies that f is crop-dependent on $\mathcal{A}$.

The converse does not hold: κ sums signed contributions, so alignment and anti-alignment may cancel, and $\kappa = 0$ does not certify crop-invariance. Attributing a positive κ specifically to the training-label construction needs the matched control of Section 3.3. So the exact zero is an algebraic identity and a check on the code, a nonzero κ is directional, and our primary comparative conclusions use the treated arm minus its matched control.

3.3. Why the measured control is not zero

A convolutional network reading crops is not a crop-invariant predictor. Zero padding at the crop border, receptive fields truncated at the window edge, and any normalisation computed over the crop all make $f(p \mid c)$ depend on c near boundaries. This is not our observation: convolutional layers are known to read absolute position from padding [19, 24] and to develop border artefacts because of it [1]. We measure the consequence directly as Ω, the probability that two crops covering the same pixel predict differently. Writing P for the pixels with $m(p) \geq 2$,

$$\Omega = \frac{\sum_{p \in P} \sum_{c \neq c' \in C(p)} \mathbf{1}[\, f(p \mid c) \neq f(p \mid c')\,]}{\sum_{p \in P} m(p)\big(m(p) - 1\big)}. \tag{8}$$

The inner sum is over ordered crop pairs (c, c'); the denominator therefore has $m(p)(m(p) - 1)$ terms for pixel p. It sums over every covered pixel rather than over $\mathcal{A}$, since it is a property of the model and is defined where the labels agree, and weights crop pairs uniformly rather than source pixels. It is shift consistency [39] on overlapping windows, and reads 0.031 for optical-only sea-ice models on repaired labels and 0.0108 for flood models on the published ones.

How far this lifts κ off zero depends on how crop-invariant the labels really are, and the two benchmarks answer differently. On flood labels regenerated from the published per-event threshold, which are crop-invariant exactly, a trained network reads $\kappa = -0.0008$, empirically near the theoretical zero. On sea ice the optical-only repaired control reads $+0.0097$. That is consistent with the labels still inheriting a per-scene normalisation a 128×128 window can partly recover, though we have not isolated that from the other crop effects above.

So the floor is not convolution alone: Ω is positive on both benchmarks while κ reads -0.0008 on one and $+0.0097$ on the other, which a resolution floor could not do. Crop-invariant training labels do not on their own hold κ at zero, and Proposition 1 does not say they do: it constrains predictors, not labels. The floor is an interaction of labels, model, crop grid and imagery, which is why every attribution claim below uses a contrast between a crop-dependent arm and its matched control on the same grid; raw values are shown to expose the floor.

4. Verifying the estimator, and bounding its scope

4.1. A case with a known answer

Proposition 1 supplies something rarer than a null: an input whose correct output is known before looking, which gives a direct test of the code. A per-pixel threshold on chip-level smoothed VH reads one pixel value and cannot see a window, so it is exactly crop-invariant. The estimator must return zero.

Measured on four held-out flood events, on the same artefact set as every other arm and between 9.5 and 41 million deterministic crop-pixel reads per event, $|\kappa|$ stays below 1.4×10^{-15} and Ω is exactly zero in all four. What remains is accumulated floating-point rounding.

An error in the $(m - 1)$ normaliser, the own-versus-other bookkeeping or the artefact-set mask surfaces here as a nonzero number. Not every error does: the weighting and lifecycle faults can pass this arm, so the released suite tests them separately with a nonuniform-coverage exact-value case and lifecycle regressions.

4.2. Which labelling rules can produce the artefact

The artefact set depends only on a labelling rule and a crop grid, not on any model, so it can be computed on imagery we did not assemble and without training anything. We ran six rules used in SAR water mapping over 445 of the 446 released Sen1Floods11 chips, one having no valid backscatter, applying each rule separately inside every crop of a 128×128 stride-32 grid and measuring $P(\mathcal{A})$: the share of covered source pixels whose valid covering-crop labels disagree.

The 17.58% here and the 24.71% calibration reference in Tab. 2 are not interchangeable prevalence estimates. This survey fits Otsu to raw VH, skips failed crops and averages chip-level fractions over 445 chips. The calibration uses a 9×9-smoothed field with finite fallbacks on a common 440-chip set, pools source pixels within each event, then averages the eleven event fractions.

Two further surveys. The four crop-adaptive rules in Tab. 1 fit a threshold or normalisation inside each crop.

labelling rule	fit inside crop?	mean $P(\mathcal{A})$	max
fixed global constant	no	0.00%	0.00%
Otsu, computed per scene	no	0.00%	0.00%
Otsu, computed per crop	yes	17.58%	40.90%
k-means, per crop	yes	17.65%	40.70%
45th percentile, per crop	yes	16.02%	32.49%
min-max per crop, then fixed	yes	12.54%	54.82%

Table 1. Artefact set produced by six labelling rules on 445 public Sen1Floods11 chips, 169 crops each. Mean and maximum are over chips after excluding the one chip with no valid backscatter. No model is involved. A rule with $P(\mathcal{A}) = 0$ leaves κ undefined rather than zero: there is nothing to average over.

Labels built from foundation-model masks do the opposite, fixing the threshold and recomputing the segmentation: SAM run inside each crop of 37 chips reads nonzero on all 37, median 1.13% and mean 8.80%, up to 43.64%, where the same regions computed once per scene read exactly 0.00%. This is an out-of-distribution stress test of natural-image SAM on duplicated SAR channels; we do not interpret it as a prevalence estimate. A third moves to 342 public CloudSEN12+ patches [3] and cloud masking. On a 128×128, stride-32 grid (144 crops per 509×509 patch), the same clear-sky percentile fitted per crop instead of per scene reads 26.57% and loses 0.079 of balanced accuracy against that dataset's expert masks. Accuracy is reported there because $P(\mathcal{A})$ counts whether covering crops agree, not whether they are right, so a labeller calling everything cloud would score a perfect zero.

The zeros are exact, on every chip and in all three surveys. Otsu is the same estimator in rows two and three of Tab. 1 and differs only in the pixels it is computed over. Applying a labeller per crop is therefore not what creates the hazard. Context-sensitive computation inside the crop is, whether a threshold, a normalisation, a background estimate or the segmentation itself, and that is a narrower and more actionable thing to check for.

4.3. κ is relative to the crop grid

How densely crops are sampled is a choice we made, so we thinned the evaluation grid from stride 32 to stride 64 on identical models and labels, recomputing nothing. The contrast rises from +0.2666 to +0.4587, a factor of 1.72, while the two geometries still order the eleven events at Spearman $r = +0.918$: the magnitude moves and the ranking survives. The supplement gives the mechanism.

The encoder moves it little. Both ends rerun with a transformer encoder and with a second convolutional one stay positive in 11 of 11 events, at +0.2966 and +0.2943 against +0.2666 for the U-Net on seed 42, with trained-control values empirically near zero in all three. The sign and rough size survive; both replacements read about 10% above the U-Net. We report this gap without an equivalence test. The trained controls are empirically near zero; only the crop-invariant predictor in Proposition 1 has an exactly zero theoretical value.

So κ compares arms sharing one crop geometry: its magnitude does not travel between papers that tile differently, and without a crop size and stride it cannot be read.

5. Experimental setup

The supplement gives the full configuration. The sea-ice benchmark has 169,051 patches over 39 tiles and 17 acquisitions; the flood benchmark has 446 chips over 11 events, and the crop sweep draws 24 crops per chip per epoch from a 13×13 grid, evaluating κ on all 169 positions. The sea-ice model carries a second branch that encodes ten co-registered ICESat-2 photon features [27] (Fig. 2). The primary κ arm, at seed 42, disables that branch and is optical-only. A seed-7 photon-enabled run is reported separately as an input-mode sensitivity check, not averaged as a repeat seed.

Splits are leave-one-acquisition-out and leave-one-event-out, since that is the unit sharing a labeller constant, and model selection never sees the held-out unit. For crop-label alignment, both scorers retain every crop-read prediction and use the source-pixel weighting, so each source pixel in the artefact set has equal weight. Only the flood expert-label analysis mosaics hard predictions by majority vote before scoring each source pixel once.

Training used two RTX A6000 cards capped at 300 W; flood crop runs take about four minutes each. Full optimisation settings are in the supplement.

6. Calibrating the diagnostic

The sea-ice benchmark cannot establish sensitivity, because the crop-dependence of that pipeline is the unknown being estimated. So we built a labeller with a dial. Sen1Floods11 [5] ships an Otsu-on-VH [30] label set with one published threshold per flood event, whose preprocessing we recover in Section 8.1. Holding that smoothing fixed and recomputing only the threshold inside each crop reproduces the sea-ice mechanism with one scalar per crop as the entire source of crop-dependence:

$$\tau_c(\alpha) = (1-\alpha)\,\tau_{\text{event}} + \alpha\,\tau_{\text{crop}}. \qquad (9)$$

At $\alpha = 0$ the labels are the published ones, crop-invariant by construction. At $\alpha = 1$, Otsu returns a crop threshold at 72,029 of 75,374 positions (95.56%);

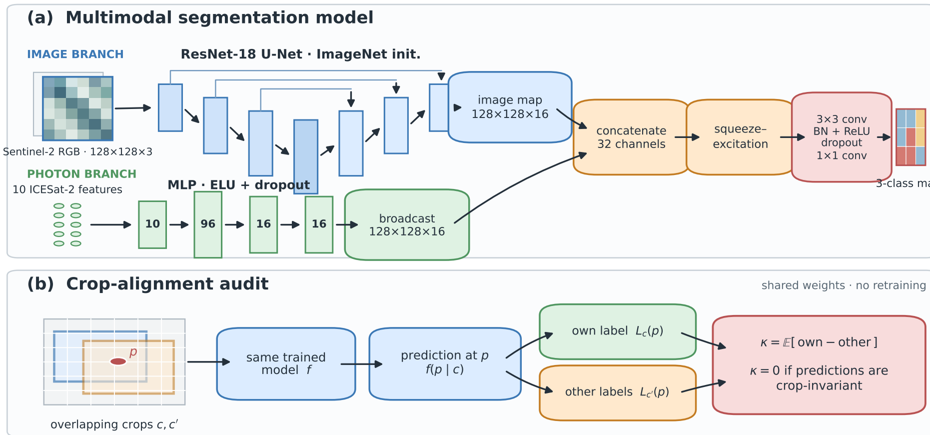

Figure 2. Model and audit schematic. The sea-ice model uses an ImageNet-initialized ResNet-18 U-Net image branch and, in photon-enabled runs, an MLP branch for ICESat-2 features. The audit keeps weights fixed and compares agreement with the current crop's label against other crops' labels for the same source pixel.

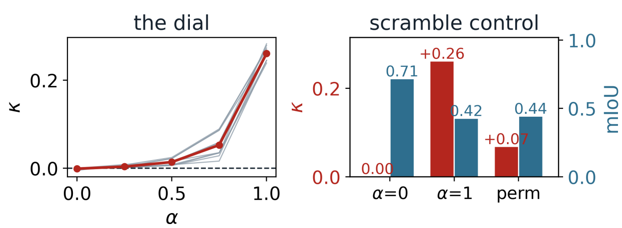

Figure 3. Left: κ against the dial of Eq. (9), grey lines held-out events and red the mean; the dashed line marks the algebraic zero. $\alpha = 0$ is the published threshold, while $\alpha = 1$ is predominantly crop-fitted (95.56% direct fits; the remainder is accounted for in text). Right: those arms and scramble, one random reassignment of threshold positions within each chip; κ red on the left axis, mIoU against expert labels blue on the right. It preserves the threshold multiset but does not eliminate spatial dependence.

	dial, α					controls	
	0.00	0.25	0.50	0.75	1.00	scramble	offset
κ	-0.0008	$+0.0037$	$+0.0141$	$+0.0528$	$+$**0.2605**	$+0.0672$	-0.0003
Ω	0.0108	0.0132	0.0211	0.0454	0.1119	0.0933	0.0121
mIoU	0.714	0.696	0.673	0.604	**0.423**	0.440	0.715

Table 2. Crop-label alignment, model crop-dependence and accuracy against expert labels, averaged descriptively over the eleven held-out events. Every κ uses the same $\alpha = 1$ reference set, with $P(\mathcal{A}^\star) = 24.71\%$. Scramble randomly reassigns thresholds within each chip; offset is constant within a chip and varies only between chips.

2,331 (3.09%) use a finite chip fallback, while 1,014 positions in six chips (1.35%) remain unusable and those chips are excluded from every arm. The within-chip range averages 6.13 dB. The endpoint also shifts the mean threshold from -21.798 to -17.796 dB. The dial controls crop fitting but does not isolate crop-dependence.

The artefact set and the labels it is scored against are both fixed at $\alpha = 1$: write $\kappa_{L^\star,\mathcal{A}^\star}(f_\alpha)$ for the model trained at α read against that fixed reference. Rebuilding either per arm would empty $\mathcal{A}$ at $\alpha = 0$, leaving the control with no pixels to be null on. So κ there is an empirical near-zero from a model that never learned the artefact; it is separate from the proposition's identity. Sea ice works the same way.

The three diagnostics in Tab. 2 answer different questions. The fixed $P(\mathcal{A}^\star)$ is a label-only prevalence and cannot rank these models; Ω is unsigned prediction inconsistency and remains positive for both near-zero controls. Only κ records the signed association between a prediction and the label of the particular crop being read.

Tab. 2 and Fig. 3 give the result. The contrast between the ends is $+0.2613$ (sd 0.0149), positive in 11 of 11 events after averaging seeds within event, and per-event rank correlation with α is perfect in all eleven. At $\alpha = 0$, where the labels are the published ones and crop-invariant exactly, κ reads -0.0008.

What changes across the dial. A threshold recomputed on a water-poor 128×128 window is not only crop-dependent, it is worse: accuracy against expert labels falls by 0.29 mIoU across the dial. None of the dial values isolates crop adaptation from the accompanying changes in threshold and label quality. The scramble arm is a sensitivity check and does not isolate causality. Within each chip it preserves the ex-

contrast	seed 7	seed 42	seed 123
α=1 against α=0	+0.2601	+0.2666	+0.2573
α=1 against scramble	+0.1992	+0.1919	+0.1888

Table 3. Training-seed sensitivity of the dial. Each cell is the mean over the same eleven held-out events, so the exact ranges across seeds are 0.0093 and 0.0104 against effects of +0.26 and +0.19.

act threshold multiset, but the unconstrained random permutation leaves 461 of 75,374 assignments (0.61%) fixed and pairs 21.78% of recipients with a geometrically overlapping donor crop; all donors also share the same chip. These geometry counts describe the full serialized mapping, including the six chip rows excluded from training and scoring. In this one reassignment, model performance retained 94% of the endpoint mIoU difference (-0.2736 versus -0.2905), while κ fell from $+0.2605$ to $+0.0672$, leaving 26% of the endpoint excess. That residual cannot be partitioned among generic crop-varying labels, overlap, and same-chip predictability. The offset arm, varying the threshold once per chip at the same spread, yields -0.0003.

Dependence of fold summaries. The eleven events are distinct, but the eleven models are leave-one-event-out folds sharing nearly all of their training data, so a t or a sign test over them assumes an independence the design does not provide, exactly as on sea ice. Table 3 offers a training-seed sensitivity check instead: the aggregate contrasts vary by less than 0.011 across three seeds. This shows stability on the same folds. It does not provide independent replication or reassignment uncertainty because all three seeds use the same scramble; Table 3 lists those seed-wise contrasts.

Scope. Sen1Floods11 as published computes one threshold per event, so the crop-dependent artefact is not present in that benchmark. The imagery and the labeller are real and the crop-dependence is ours. This experiment calibrates the diagnostic; it does not establish a second real-world occurrence, and nothing here bears on that dataset's published labels.

7. Association with expert-label performance

The models of Section 6 were trained, validated and tested inside their own algorithmic label field. We additionally score each model against the expert labels that Sen1Floods11 ships, without new training. A mosaic majority vote gives one decision per source pixel before comparison; patch scoring would count each pixel about sixteen times and reward the crop-specific target repeatedly.

Expert-label mIoU decreases monotonically from 0.714 to 0.423 across the constructed dial, an endpoint difference of -0.2905 (descriptive sd 0.0918), with the same direction in all eleven held-out events. This is the combined result of replacing published event thresholds with predominantly per-crop Otsu labels. It is not the causal cost of crop-dependence: the endpoint also moves the mean threshold by 4.00 dB and changes class balance, boundaries and raw label quality.

After subtracting each event's mean, κ and expert-label mIoU have descriptive $r = -0.928$ and a slope of -1.05 mIoU per unit κ across the five dial levels. Because α jointly drives both axes, this is a joint dose response, not evidence that κ predicts damage beyond the dial. Fixed-α correlations do not give that stronger result: their average is -0.196, with signs ranging from $+0.774$ at $\alpha = 0$ to -0.087 at $\alpha = 1$.

The within-chip scramble gives a limited sensitivity check. In its one threshold reassignment, 94% of the endpoint mIoU difference remained while the alignment fell to 26%. Since the mapping allows fixed points, overlapping donor crops and same-chip spatial dependence, we do not interpret that contrast as isolating a mechanism.

8. Applying and reconstructing two published labellers

8.1. What we recovered, and on whose data

Provenance, and which step is the culprit. We are explicit about what is ours, because it bears on how far this generalises. The colour labeller is published [17, 18], and so is the practice of cutting Sentinel-2 patches on ICESat-2 track points one pixel apart, which produces densely overlapping windows [26]. The 17-acquisition assembly is ours: our co-location, our 128×128 geometry, our class-stratified subsample.

The published colour thresholds are fixed constants, and the released code applies them after splitting a scene into patches. What makes our label set crop-dependent is one stage earlier, a cloud and shadow removal that fits three quantities to the array it is handed: an Otsu threshold, a median-blur background estimate, and two min-max normalisations. Labelling per crop is not the hazard, since a fixed rule survives it. Context-sensitive computation inside the crop is what creates it, though not every fitted step contributes, and the experiments below identify which ones do.

External imagery. We tested the released labeller on the 446 public Sen1Floods11 optical chips on the same stride-32 grid. Nothing is trained and input scaling is one global constant, so any $P(\mathcal{A}) > 0$ certifies crop dependence of the rule under this fixed preparation; the class names do not describe flood scenes and need not, since $P(\mathcal{A})$ measures agreement among crops without judging correctness. Two arms serve as code controls: stage 1 alone, and the stage with every fitted quantity taken from the scene, are compositions of per-pixel functions and must read zero. Both do, exactly. Fitted per crop as released the stage reads 37.04%. Repairing its background and Otsu threshold cuts that by 88% to 4.50%, but the worst chip falls only from 57.90% to 40.07%, and the two normalisations, inert while the other quantities are fitted per crop, carry what is left: a partial repair is not a repair. The supplement ablates each step and varies the one preparation constant we chose, which the residual is sensitive to and the 37.04% is not.

Sea ice. OpenCV hue ends at 179, below the published upper threshold of 185, so only the value channel acts; a grid search recovers the water constant in all 17 folds. Crop-dependence enters through the cloud and shadow stage, run here on 128×128 patches with a 155-pixel median kernel. Scene-wide regeneration raises crop unanimity from 0.9678 to 1.0000 in every fold.

Floods. Sen1Floods11 ships an expert label and an Otsu label for the same 446 chips, and unlike the sea-ice case it publishes the constant, so the recovery can be checked from outside. Thresholding the raw band leaves a bias of one sign on all eleven events, a systematic signature, and the supplement sweeps the smoothing radius to find it: a 9×9 focal mean before thresholding lands all eleven within 0.25 dB at 99.08% agreement. The constant moves 5.57 dB across events, so no global threshold can express it and a model seeing the whole chip can infer which event it is in and adapt.

8.2. Crop-label alignment on the sea-ice benchmark

Sea-ice crops overlap about 33 times, so a source pixel is seen inside many crops, and under the per-crop scheme those crops can disagree. Across all 17 folds, with $\mathcal{A}$ averaging 13.22% of covered pixels and 23.9M deterministic crop-pixel reads per fold, the model trained on crop-noisy labels in the optical-only primary arm (seed 42) reads $\kappa = +0.1518$. Its matched control, trained on repaired labels and evaluated on identical pixels, reads +0.0097. The mean paired difference is **+0.1421**, and it is positive on each of the 17 dependent evaluation folds. A photon-enabled sensitivity at seed 7 reads a +0.1454 contrast, also positive in 17 of 17 folds. Because input mode and initialization change together, this serves only as a robustness check because it cannot estimate seed replication. Figure 4 gives the optical-only primary per-fold results against their controls.

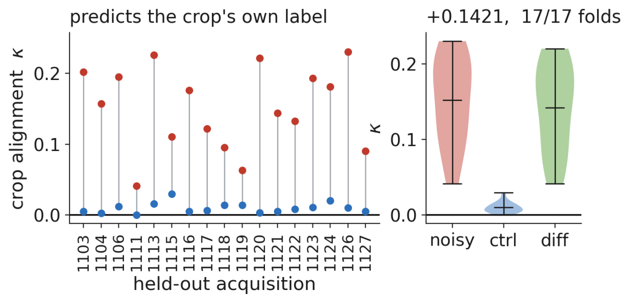

Figure 4. Optical-only primary crop-label alignment (seed 42) per held-out acquisition. Red is the model trained on crop-noisy labels, blue its matched control trained on labels regenerated once per scene. Zero is the theoretical identity for crop-invariant predictors; the trained control is empirical and need not equal it.

We report that descriptively and attach no test to it. These 17 folds are overpasses of 39 tiles across 25 days sharing tiles, geography and almost all of their training data, so a t or an interval over them would assume an independence the design does not provide. We therefore omit inferential summaries over folds.

We do not express this as a fraction of the calibrated contrast of Section 6: the two used different crop grids, and Section 4.3 shows a grid change moves the contrast by more than any such ratio would report. Two consistency checks follow. On the repaired label set $\mathcal{A}$ is empty, since crop unanimity of 1.0000 means every covering crop agrees. And Ω is exactly zero for the two-parameter threshold, whose value channel is a per-pixel function of per-fold constants.

9. Discussion

Benchmark builders should fit contextual quantities on support containing all evaluation crops that cover a source pixel, persist one label per source pixel, split by the labeller's dependence unit, score each source pixel once, and report a context-free baseline. Reviewers should ask for κ on a stated crop grid with a matched control. The algebraic zero checks the diagnostic; the control is needed for attribution.

The diagnostic is narrow: it applies only where overlapping crops disagree, and it misses crop-invariant pseudo-label shortcuts, calibration failures and unrelated shortcuts. The calibration dial is constructed, sea ice lacks human reference labels, and folds are dependent.

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